



Consumer Goods Analytics

Code-basic Resume Project Challenges # 4

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AGENDA

ATLIQ BUSINESS MODEL

PROBLEM STATEMENT AND
PROJECT OVERVIEW

DATA-SET

10-AD-HOC REQUEST

ABOUT ATLIQ

→ AtliQ Hardware, is a leading hardware company specializing in PCs, Printers, mouse and computers with a global reach

PROBLEM STATEMENT

→ AtliQ Hardware, a leading computer hardware Producer, faced a critical challenge

→ They needed quick and data-informed decisions to stay competitive in the ever-evolving market.

→ The management noticed that they were missing crucial insights for strategic moves

PROJECT OVERVIEW

→ In this Project, I will be working with a data-set related to consumer goods. The goal is to answer ten specific AD HOC REQUEST using SQL queries.

ATLIQ HARDWARE-BUSINESS MODEL

CUSTOMER CHANNELS

CUSTOMER

CUSTOMER PLATFORMS

CROMA

BEST BUY

STAPLES

FLIPKART

BRICK AND
MORTAR

E-COMMERCE

FLIPKART

CROMA

BEST BUY

AMAZON

Retailer

cromā
The Electronics Megastore

amazon

Direct

AtliQ
HARDWARE
e store

AtliQ
HARDWARE
exclusive

Distributor

NEPTUNE

ATLIQ HARDWARE-BUSINESS MODEL

PRODUCT DIVISION

PC

N&S

P&A

WIFI
EXTENDERS

USB FLASH
DRIVE

PROCESSOR

MOTHERBOARD

PERSONAL
LAPTOP

GAMING
LAPTOP

KEYBOARD

MOUSE

PRODUCT SEGMENT

STORAGE

PERIPHERALS

NOTEBOOK

NETWORKING

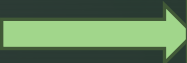
DESKTOP

ACCESSORIES



DATA-SET

DATABASE-GDB041



TABELS



DIM CUSTOMER

DIM PRODUCT

FACT GROSS PRICE

FACT SALES MONTHLY

FACT MANUFACTURING COST

FACT PRE INVOICE DEDUCTION

AD_HOC-REQUEST: 1



QUESTION:

Provide the list of market in which customer “AtliQ Exclusive” Operates its Business in the APAC region

SQL QUERY

```
SELECT DISTINCT (market)
FROM dim_customer
WHERE customer = "AtliQ Exclusive"
AND region = "APAC"
```

Result Grid	
	market
▶	India
	Indonesia
	Japan
	Philippines
	South Korea
	Australia
	Newzealand
	Bangladesh



INSIGHTS

- AtliQ Exclusive has a presence in several countries across the APAC region, including India, Indonesia, Japan, Philippines, South Korea, Australia, New Zealand, and Bangladesh
- It reflects a robust market presence and adaptability to different cultural and economic contexts in the Asia-Pacific region.



QUESTION

What is the percentage of unique product increase in 2021 vs. 2020?
The final output contains these fields, unique_products_2020
unique_products_2021 Percentage chg

SQL QUERY

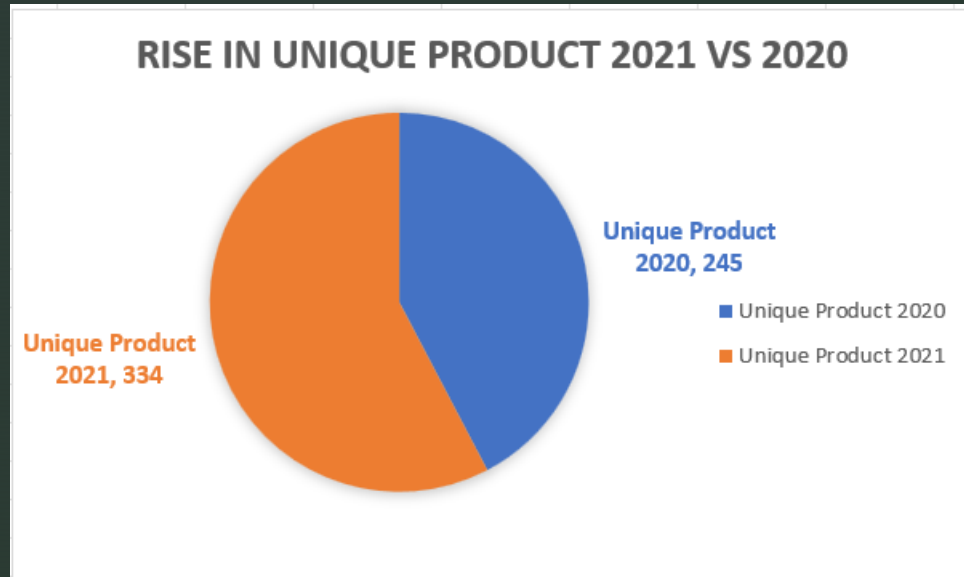
```
With x AS
(SELECT count(DISTINCT product_code) AS uniq_Product_2020
FROM fact_sales_monthly
WHERE fiscal_year = 2020),
y AS ( SELECT count(DISTINCT product_code) AS uniq_product_2021
FROM fact_sales_monthly
WHERE fiscal_year = 2021)

SELECT
x.uniq_product_2020 AS x,
y.uniq_product_2021 AS y,
ROUND(
((y.uniq_product_2021-x.uniq_product_2020)/x.uniq_product_2020)*100,2) AS PCT_Chng
FROM x,y
```

Result Grid Filter Rows:			
	x	y	PCT_Chng
▶	245	334	36.33



VISUAL



INSIGHTS

- There was a significant Increase in Unique Product with 334 In 2021 Compared to 245 in 2020
- The percentage change represents a growth of 36.33% In Unique Products from one year to the next
- This substantial Increase in unique Products suggest a Focus on expanding product offering which can attract a boarder customers base and potentially boost sales and revenue
- it's a positive indicator of business growth and adaptability to change market demands



QUESTION

Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields, Segment Product Count

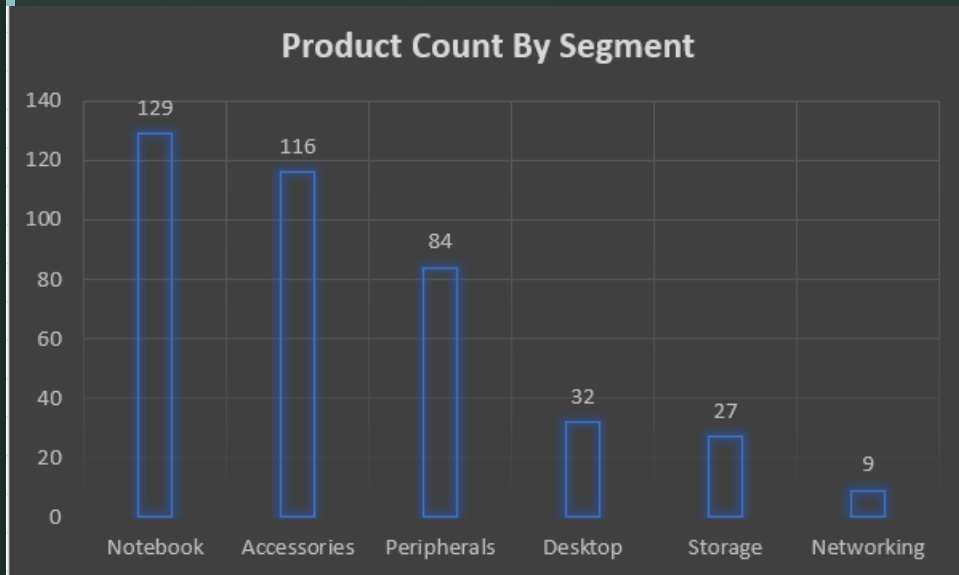
SQL QUERY

```
SELECT segment,  
count(DISTINCT product_code) AS UNIQ_product  
FROM dim_Product  
GROUP BY segment  
ORDER BY UNIQ_product DESC
```

Result Grid Filter Rows:		
	segment	UNIQ_product
▶	Notebook	129
	Accessories	116
	Peripherals	84
	Desktop	32
	Storage	27
	Networking	9



VISUAL



INSIGHTS

- The "Notebook" segment has the highest product count, with 129 products.
- The "Networking" segment has the smallest product count with only 9 products.
- These insights indicate the diversity and product offerings within each segment, with "Notebook" and "Accessories" offering a wide range of options, while "Networking" has a more limited selection.
- The variety in product offerings allows catering to various customer preferences and needs across different segments



QUESTION

Follow-up: Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields, segment Product Count 2020 Product Count 2021 difference

SQL QUERY

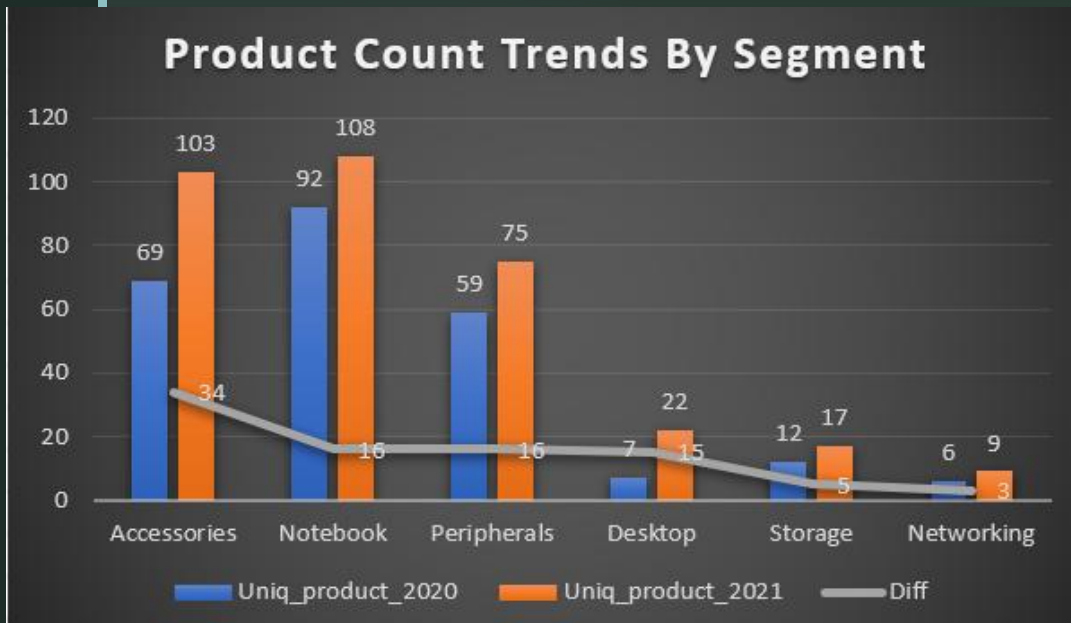
```
With x AS
( SELECT p.segment, count(DISTINCT Product_code) AS Uniq_product_Cnt_2020
  FROM dim_product AS p
JOIN
fact_sales_monthly AS f
USING (Product_Code)
WHERE f.fiscal_year = 2020
GROUP BY p.segment ),
y AS
( SELECT p.segment, count(DISTINCT Product_code) AS Uniq_product_cnt_2021
  FROM dim_product AS p
JOIN
fact_sales_monthly AS f
USING (Product_Code)
WHERE f.fiscal_year = 2021
GROUP BY p.segment )

SELECT x.segment, Uniq_product_cnt_2020,
Uniq_product_cnt_2021, ABS(x.Uniq_product_cnt_2020-y.Uniq_product_cnt_2021) AS Diff
FROM x JOIN y ON x.segment=y.segment
ORDER BY diff DESC
```

segment	Uniq_product_cnt_2020	Uniq_product_cnt_2021	Diff
Accessories	69	103	34
Notebook	92	108	16
Peripherals	59	75	16
Desktop	7	22	15
Storage	12	17	5
Networking	6	9	3



VISUAL



INSIGHTS

- The "Accessories" segment saw a significant increase in product counts, with 34 more products in 2021 compared to 2020.
- These insights suggest a general trend of product portfolio growth, particularly in the "Accessories" and "Notebook" segments, which can indicate a strategy to cater to a broader range of customer preferences.
- The growth in product variety may enhance market competitiveness and offer customers more choices.



QUESTION

Get the products that have the highest and lowest manufacturing costs.
The final output should contain these fields, product code, product
Manufacturing cost

SQL QUERY

```
SELECT m.product_code, P.Product, M.manufacturing_cost
FROM fact_manufacturing_cost AS m
JOIN dim_product AS p
USING (Product_code)
WHERE m.manufacturing_cost = (SELECT max(manufacturing_cost)
FROM fact_manufacturing_cost) OR
m.manufacturing_cost = (SELECT min(manufacturing_cost) FROM fact_manufacturing_cost)
ORDER BY m.manufacturing_cost DESC;
```

product_code	Product	manufacturing_cost
A6121110208	AQ HOME Allin1 Gen 2	263.4207
A2118150101	AQ Master wired x1 Ms	0.8654

INSIGHTS

- "AQ HOME Allin1 Gen 2" has a relatively higher manufacturing cost of 240.5364.
- In contrast, "AQ Master wired x1 Ms" has a significantly lower manufacturing cost of 0.892.



QUESTION

Generate a report which contains the top 5 customers who received an average high Pre Invoice Discount Pct for the fiscal year 2021 and in the Indian market. The final output contains these fields, Customer Code Customer Average Discount Percentage

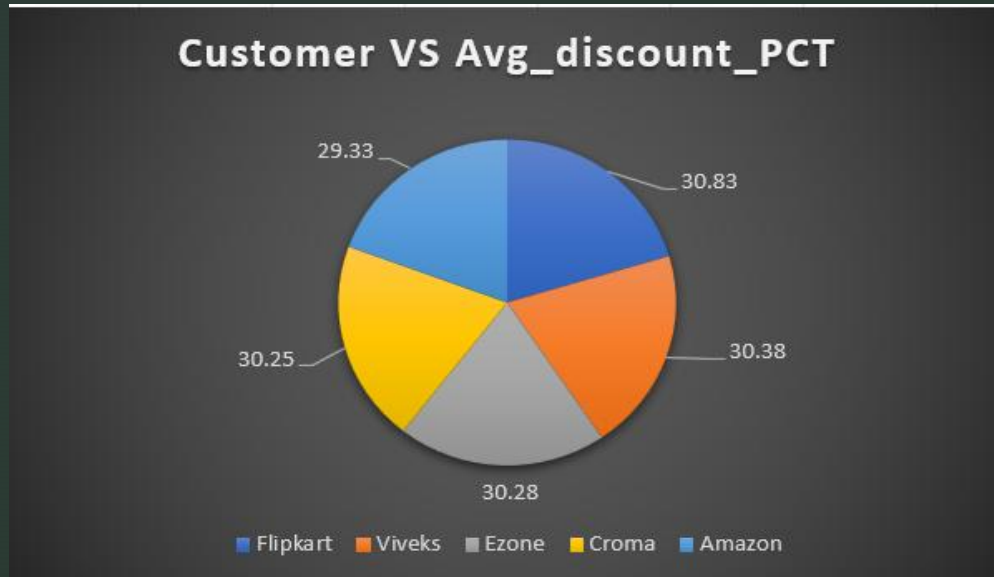
SQL QUERY

```
SELECT Pre.customer_code,c.customer,  
ROUND(Avg(pre.pre_invoice_discount_PCT)*100,2) AS Avg_discount  
FROM pre_invoice_dis_pct AS Pre  
JOIN  
    dim_customer AS c  
    USING(customer_code)  
WHERE fiscal_year = 2021 AND c.market = 'india'  
GROUP BY Pre.customer_code, c.customer  
ORDER BY Avg_discount DESC  
LIMIT 5;
```

Result Grid				Filter Rows:
	customer_code	customer	Avg_discount	
▶	90002009	Flipkart	30.83	
	90002006	Viveks	30.38	
	90002003	Ezone	30.28	
	90002002	Croma	30.25	
	90002016	Amazon	29.33	



VISUAL



INSIGHTS

- Flipkart has the highest average pre-invoice discount percentage at 30.83%.
- Amazon has a comparatively lower average discount at 29.33%.
- These insights indicate varying discount strategies among customers, with Flipkart and Viveks offering the highest average discounts, potentially attracting cost-conscious shoppers.
- Amazon provides relatively lower discounts, suggesting a different pricing strategy or a customer base less sensitive to discounts. These insights can be valuable for adjusting discount strategies and understanding customer preferences.



QUESTION

Get the complete report of the Gross sales amount for the customer “AtliQ Exclusive” for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions. The final report contains these columns: Month Year Gross sales Amount

SQL QUERY

```
SELECT monthname(s.date) AS Months,s.fiscal_year,
ROUND(SUM(s.sold_quantity*g.gross_price),2) AS Gross_Price_Total
FROM fact_sales_monthly AS s
JOIN dim_customer AS c
USING (Customer_code)
JOIN fact_gross_price AS g
USING (product_code)
WHERE c.customer = "AtliQ Exclusive" AND s.Fiscal_Year = 2020
GROUP BY monthname(s.date), s.Fiscal_Year
ORDER BY s.Fiscal_Year
```

```
SELECT monthname(s.date) AS Months,s.fiscal_year,
ROUND(SUM(s.sold_quantity*g.gross_price),2) AS Gross_Price_Total
FROM fact_sales_monthly AS s
JOIN dim_customer AS c
USING (Customer_code)
JOIN fact_gross_price AS g
USING (product_code)
WHERE c.customer = "AtliQ Exclusive" AND s.Fiscal_Year = 2021
GROUP BY monthname(s.date), s.Fiscal_Year
ORDER BY s.Fiscal_Year
```

Months	fiscal_year	Gross_Price_Total
September	2020	17040562.24
October	2020	19475069.34
November	2020	28512004.15
December	2020	18322529.10
January	2020	18009123.49
February	2020	15171931.84
March	2020	1422525.16
April	2020	1492369.18
May	2020	2971173.85
June	2020	6451963.41
July	2020	9685828.63
August	2020	10599401.49

Months	fiscal_year	Gross_Price_Total
September	2021	37752848.19
October	2021	40444450.56
November	2021	62302295.57
December	2021	39306619.91
January	2021	37704996.30
February	2021	30852326.32
March	2021	36972600.02
April	2021	22147394.99
May	2021	37037201.24
June	2021	37037201.24
July	2021	36677913.91
August	2021	21839400.72

INSIGHTS



- November 2021 had the highest gross sales amount, reaching \$32,247,289.79.
- In contrast, the fiscal year 2021 started with lower sales in September but still had a significant peak in November.
- There is a notable seasonality in sales, with November being a consistently strong month.
- The months of March and April in fiscal year 2020 had relatively low sales, which improved in fiscal year 2021.
- These insights can guide strategic decisions, such as focusing marketing efforts and inventory planning around the peak sales months and addressing potential challenges during lower sales months.



QUESTION

In which quarter of 2020, got the maximum Total Sold Quantity? The final output contains these fields sorted by the Total Sold Quantity, Quarter Total Sold Quantity

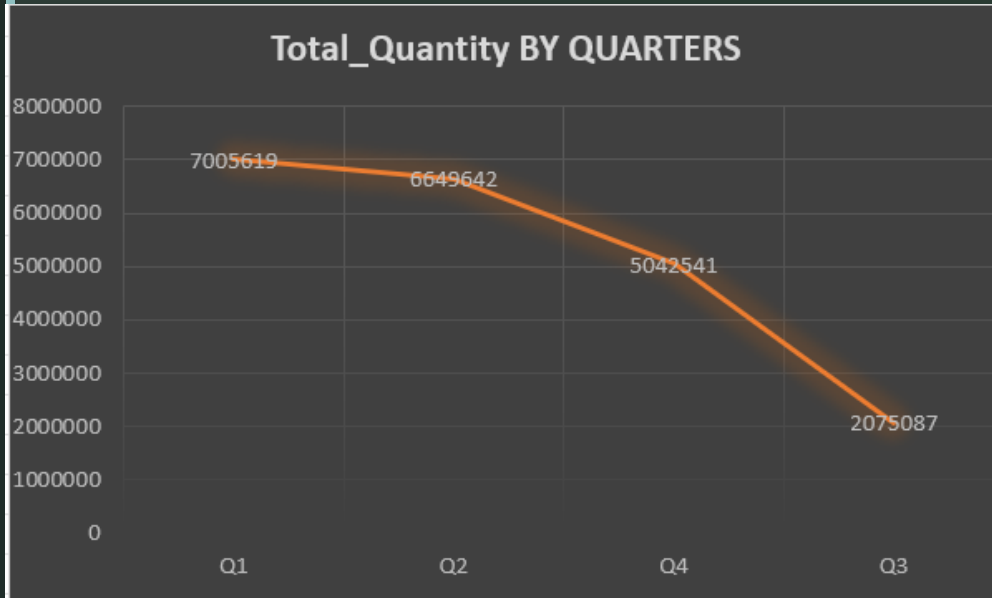
SQL QUERY

```
SELECT
CASE
    WHEN month(date) IN (9,10,11) THEN "Q1"
    WHEN month(date) IN (12,1,2) THEN "Q2"
    WHEN month(date) IN (3,4,5) THEN "Q3"
    ELSE "Q4"
END AS Quarters,
SUM(sold_quantity) AS Total_Quantity
FROM fact_sales_monthly
WHERE Fiscal_Year = 2020
GROUP BY Quarters
ORDER BY Total_Quantity DESC;
```

Result Grid   Filter Rows:		
	Quarters	Total_Quantity
▶	Q1	7005619
	Q2	6649642
	Q4	5042541
	Q3	2075087



VISUAL



INSIGHTS

- The highest total sold quantity is in Q1, with 7,005,619 units.
- These insights highlight a seasonal variation in sales, with Q1 and Q2 being the strongest quarters and Q3 being the weakest.
- This information is valuable for planning inventory and marketing strategies to align with seasonal demand



QUESTION

Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields, channel
Gross Sales Millions Percentage

SQL QUERY

```
WITH x AS
( SELECT c.channel,
  ROUND(SUM(s.sold_quantity*g.gross_price)/1000000,2) AS Gross_Sales_Mln
  FROM fact_sales_monthly AS s
  JOIN dim_customer AS c
  USING (customer_code)
  JOIN fact_gross_price AS g
  USING (Product_code)
  WHERE s.Fiscal_Year =2021
  GROUP BY c.channel )

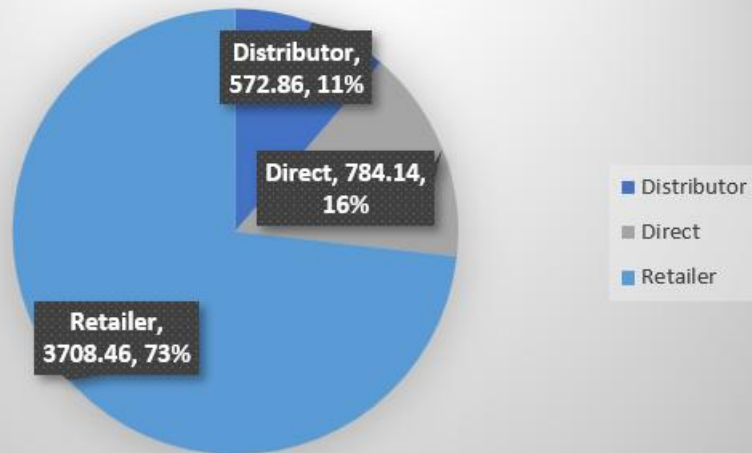
SELECT channel, Gross_Sales_mln,
ROUND((Gross_sales_mln/(SELECT SUM(Gross_sales_Mln) FROM x))*100,2) AS PCT
FROM x
ORDER BY Gross_Sales_mln
```

Result Grid			
Filter Rows:			
	channel	Gross_Sales_mln	PCT
▶	Distributor	572.86	11.31
	Direct	784.14	15.48
	Retailer	3708.46	73.21



VISUAL

Gross Sales Channel Contribution



INSIGHTS

- The "Retailer" channel accounts for the majority of sales, contributing to 73.21% of gross sales.
- The "Direct" channel also plays a significant role, representing 15.48% of gross sales.
- The "Distributor" channel contributes 11.31% of gross sales.
- A significant focus on the "Retailer" channel suggests it is the primary revenue driver. Diversification and growth opportunities may be explored in the "Direct" and "Distributor" channels to further maximize sales.



QUESTION

Get the Top 3 products in each division that have a high Total Sold Quantity in the Fiscal year 2021? The final output contains these fields, division Product code

SQL QUERY

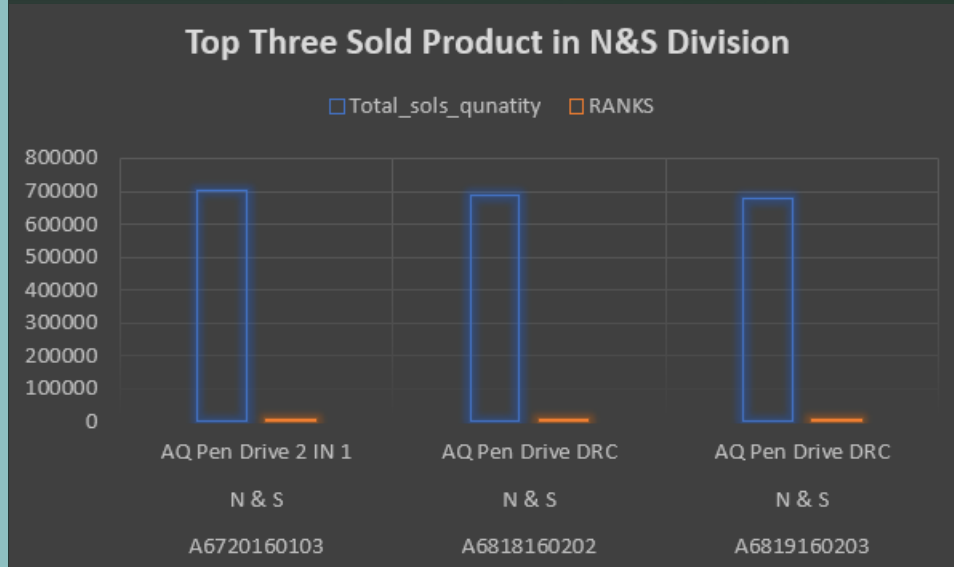
```
WITH x AS
( SELECT  p.division,p.product_code,p.product,
          SUM(s.sold_quantity) AS Total_sols_qunatity,
          Rank() OVER (PARTITION BY p.division ORDER BY SUM(s.sold_quantity) DESC ) AS RANKS
  FROM dim_product AS p
  JOIN fact_sales_monthly AS s
  USING (product_code)
  WHERE s.Fiscal_Year = 2021
  GROUP BY p.division,p.product_code,p.product )

SELECT * FROM x
WHERE RANKS IN (1,2,3)
ORDER BY division,RANKS
```

	division	product_code	product	Total_sols_qunatity	RANKS
▶	N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
	N & S	A6818160202	AQ Pen Drive DRC	688003	2
	N & S	A6819160203	AQ Pen Drive DRC	676245	3
	P & A	A2319150302	AQ Gamers Ms	428498	1
	P & A	A2520150501	AQ Maxima Ms	419865	2
	P & A	A2520150504	AQ Maxima Ms	419471	3
	PC	A4218110202	AQ Digit	17434	1
	PC	A4319110306	AQ Velocity	17280	2
	PC	A4218110208	AQ Digit	17275	3



VISUAL



INSIGHTS

- The best-selling product in the "PC" division is "AQ Digit," with two variants followed by "AQ velocity".
- The division specializes in PC-related products, indicating a focus on the computing niche.
- Efficient inventory management is essential to meet customer demand. Ongoing branding and marketing efforts are important to maintain success.

**THANK
YOU**